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Diffractive Metasurface-Enabled Single-Shot Characterization of Hybrid-Order Poincaré Beams

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Hybrid-order Poincaré beams (HyOPBs) with complex transverse polarization states hold significant potential in optical communication and quantum information. Fully characterizing the Poincaré parameters of HyOPBs is a key task to accelerate applications. Conventional methods typically require mapping in multiple polarization states and reconstructing point by point, creating a fundamental bottleneck for fast measurement and real-time monitoring of Poincaré parameters. In this work, a single-shot and real-time characterization scheme for HyOPBs is demonstrated by applying diffractive neural networks to a system of cascaded diffractive metasurfaces. The designed diffractive metasurfaces essentially function as an optical processor, efficiently extracting high-dimensional spatial modes and complex amplitude information. Whereafter, the Poincaré parameters are accurately predicted, and the HyOPBs are correctly reconstructed with the help of electronic deep neural networks. This innovative approach is validated through a series of simulation studies with average reconstruction errors of <2.68% for σ and 1.84% for θ, respectively. The work provides an effective strategy for precise, compact, and real-time detection of HyOPBs, paving the way for their application in the next generation of high-capacity optical communication systems.

1. Introduction

Hybrid-order Poincaré beams (HyOPBs), which integrate orbital angular momentum (OAM) and spin angular momentum (SAM) to represent the joint spatial evolution of phase and polarization, play a key role in diverse fields, such as optical communications, sensing, and quantum information.[1–9] Owing to their infinite OAM orthogonal states and dynamic polarization capabilities, HyOPBs have emerged as a significant focus in high-throughput optical communications.[10–12] In addition to detecting OAM orthogonal states,[13–15] the state of polarization (SOP) of HyOPBs represents a critical resource for optical communications, as it facilitates the implementation of high-throughput vector mode multiplexing.[16–19] Consequently, Single-shot and real-time characterization of the SOP is a prerequisite for advancing both classical and quantum optical communications.

Traditionally, detection methods have relied on Stokes polarimetry,[20–24] which generally requires at least four independent measurements, each projecting the state onto a different polarization component. This approach requires multiple reconfigurable polarization optical elements, leading to bulky setups with limited temporal resolution. Recent work has demonstrated a single-shot, full characterization of the complex amplitude of vector light fields using spatial Stokes tomography.[25,26] However, in all these prior experiments, the SOP characterization demands pixel-precise alignment and point-to-point calculation of the measured Stokes parameters, introducing registration errors and complicating the processing. Therefore, these methods are not well suited for accurate, real-time, and compactly integrated detection of HyOPBs.

Metasurfaces, composed of arrays of subwavelength optical scatterers, can tailor the amplitude, phase, and polarization characteristics of light fields,[25–28] providing a versatile platform for compact planar optics. To date, notable applications of metasurfaces include metalenses, holography, structured light generation, diffractive neural networks, and optical computing.[29–50] Recent advances have demonstrated the feasibility of metasurface-enabled ellipsometry for single-measurement polarization detection.[51–53] These devices operate by measuring intensity under different polarization bases and retrieving the Stokes parameters to accurately determine the polarization state. However, despite their compactness and efficiency, most of these devices are limited to resolving polarization states on a standard Poincaré sphere. For HyOPBs,

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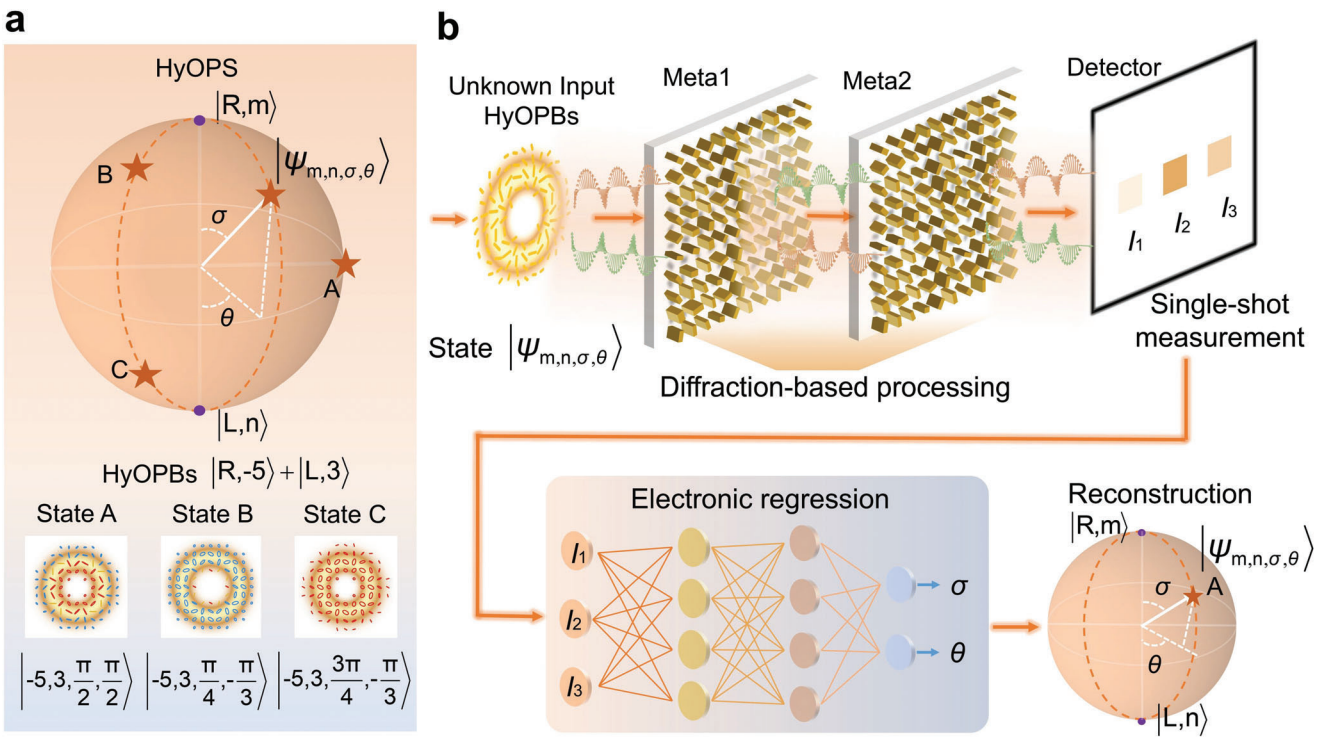

Figure 1. Diffractive metasurface-enabled single-shot HyOPBs characterization. a) Schematic diagram of HyOPBs. The poles of the Poincaré sphere correspond to LCP and RCP vortex beams, which are the orthogonal polarization bases with distinct topological charges. Each star point on the Poincaré sphere represents HyOPBs with distinct vector field distributions. The inset at the bottom displays the polarization and intensity distributions of three arbitrarily selected HyOPBs. b) Schematic diagram of single-shot HyOPBs characterization. The double-layer diffractive metasurface functions as an optical computing layer, directly processing the incident vector field and converting it into three distinct points on the focal plane. The detector is employed to capture the corresponding intensity distribution in a single measurement. Subsequently, the Poincaré parameters are fitted and reconstructed using an electronic deep neural network.

particularly those with complex vector light fields and large angular quantum numbers, metasurface-based ellipsometers struggle with accurate characterization due to the need for additional spatial multiplexing and separation. Consequently, the realization of metasurface-enabled, accurate, fast and single-shot characterization of HyOPBs has not yet been achieved and remains an area for further exploration.

In this work, we propose a direct single-shot method for the complete characterization of arbitrary HyOPBs using a spin-multiplexed diffractive metasurfaces. The metasurfaces, designed via a diffractive neural network, map the two circularly polarized waves of HyOPBs to three prescribed focal points on the output plane. Subsequently, an electronic neural network reconstructs all relevant SOP information of HyOPBs, including amplitude and phase difference, from the intensities of these focal points. This not only enables single-shot and real-time characterization, but also eliminates extra measurement errors and minimizes the device size by removing the need for reconfigurable polarization elements. In particular, our approach avoids reliance on polarization basis projections, fundamentally distinguishing it from traditional approaches that depend on Stokes parameter measurements. As a result, it eliminates the errors typically introduced by spatial registration and separation in Stokes parameter assessments. The designed diffractive metasurfaces essentially represent an optical processor supporting arbitrary vector mode transformations. These diffractive metasurfaces facilitate single-shot, real-time, and accurate characterization of HyOPBs in a compact and integrated format, paving the way for advancements in high-capacity optical communications, entangled quantum information encoding, and high-dimensional optical encryption.

2. Principles and Design

2.1. Working Principles of Single-Shot HyOPBs Characterization

The concept of HyOPBs single-shot characterization using diffractive metasurfaces are schematically illustrated in **Figure 1**. In principle, the state of HyOPBs can be represented by the linear superposition of left circularly polarized (LCP) and right circularly polarized (RCP) optical vortices with distinct topological charges[54]

$$\left|\psi_{m,n,\sigma,\theta}\right\rangle = \sin(\frac{\sigma}{2})e^{i\theta/2}\left|R,m\right\rangle + \cos(\frac{\sigma}{2})e^{-i\theta/2}\left|L,n\right\rangle \tag{1}$$

where $|R, m\rangle$ and $|L, n\rangle$ represent RCP and LCP optical vortices with different topological charges of m and n, respectively. σ and θ represent the amplitude and phase difference of the orthogonal polarization bases, corresponding to the azimuthal angle and polar angle on the hybrid-order Poincaré Sphere (HyOPS). Generally, the conventional strategy for the complete characterization of HyOPBs involves obtaining the intensity distribution

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through four independent projection measurements to recover the Stokes parameters $|s\rangle = |s_0,s_1,s_2,s_3\rangle$. The HyOPBs can then be reconstructed by calculating the Stokes parameters point by point, a process that may introduce alignment errors due to spatial or temporal division.[55,56]

In our scheme, we consider the linear superposition of vector modes to develop a more efficient approach compared to measuring the polarization state with spatial contributions. When a vector mode passes through a linear system with negligible polarization crosstalk, the global optical response is the linear superposition of two separate optical responses induced by two orthogonal circular polarization bases. In this regard, we introduce a spin-multiplexed diffractive metasurface that can independently process two circularly polarized light vortices to produce two sets of predefined outputs. Then, the synthesized output is the linear superposition of the two sets of outputs. In other words, each vector mode produces a distinct output. Therefore, it provides the possibility to directly reconstruct the Poincare parameters by measuring the output intensity.

To be specific, the introduced diffractive metasurfaces act as an optical computation layer after training, processing the incident complex light field into the latent feature space. The intensity measurements at the output positions are closely related to the complex amplitude coefficients of the orthogonal polarization bases. Therefore, by determining the implicit relationship between the intensity measurement $|I\rangle = |I_1,I_2,I_3\rangle$ and the input state $|\psi_{m,n,\sigma,\theta}\rangle$, we can accurately predict the Poincare parameters $|Q\rangle = |\sigma, \theta\rangle$of the input beam in a single characterization. Our method directly processes the full vectorial light field without additional optical elements, thereby eliminating alignment errors.

In contrast to previous diffractive optical neural networks (DONNs), the vector parameter prediction of HyOPBs is essentially a regression photonic problem. Due to the limited nonlinearity, it is quite challenging to accurately predict vector parameters merely using optical neural networks. Therefore, we introduce an electronic deep neural network (EDNN) to establish a complex mapping M: $I \rightarrow Q$ between the intensity measurement $|I\rangle$ and the reconstructed Poincaré parameters $|Q\rangle$. The EDNN with powerful feature learning and nonlinear modeling capabilities can handle complex high-dimensional data, which makes it quite suitable for data fitting in this context. Consequently, we can use the EDNN to promptly obtain the vector parameters of the HyOPBs by processing the output intensity.

2.2. Design of Spin-Multiplexed Diffractive Metasurface for Optical Computing

To fully illustrate the implementation procedure of our strategy, we take the HyOPBs = $|\psi_{-3,3,\theta,\sigma}\rangle$as an example and prepare the designed spin-multiplexed diffractive metasurfaces. The key aspect of our single-shot characterization method is mapping Poincaré parameters $|Q\rangle$, which is particularly challenging to measure directly but instead can be converted into three focus intensities $|I\rangle$. This transformation of the vector light field for HyOPBs is illustrated in **Figure 2a**. It is important to note that we selected three intensity values to map the feature space, as the Poincaré parameters are inherently defined by three distinct parameters, while the commonly used two-parameter representation arises from normalization. The three focus intensities provide sufficient redundancy to reconstruct the Poincaré parameters. According to Equation (1), the HyOPBs can be decomposed into complex amplitude superpositions of two orthogonal circularly polarized light vortices (left side of Figure 2a). We use a spin-multiplexed diffractive metasurface to independently process each orthogonal circularly polarized light vortex, and thus generate two sets of predefined outputs (middle of Figure 2a). When a vector field with varying Poincaré parameters is incident, the final combined output can be regarded as the complex amplitude superposition of the output fields from the LCP and RCP channels (right of Figure 2a). Since the predefined output is fixed, the actual output is closely linked to the vector parameters of the input field, allowing for the reverse reconstruction of the HyOPB's vector parameters by analyzing the measured output.

Specifically, for the LCP optical vortices, the target amplitudes of the three designed foci satisfy $A_1 = A_2 = A_3 = 1$, and the corresponding target phases satisfy $\varphi_1 = \frac{\pi}{2}, \varphi_2 = \varphi_3 = 0$. The principle of parameter selection is based on orthogonality, meaning the feature space is optimized by maximizing the differences in intensity among the three design foci. In our design, three fixed-size foci are positioned at distinct locations on the output plane. The intensities are assigned as $A_1 = A_2 = A_3 = 1$, and their phases are sequentially defined as $\varphi_1 = \frac{\pi}{2}, \varphi_2 = \varphi_3 = 0$, thereby establishing the target output field **U**. Similarly, for the RCP optical vortices, the target amplitudes of the three foci satisfy $A_1 = A_2 = A_3 = 1$, and the corresponding target phases satisfy $\varphi_1 = \varphi_2 = 0, \varphi_3 = \frac{\pi}{2}$. Notably, according to the principle of linear superposition, when the spatial positions of the three foci coincide, the focal amplitude detected can be expressed as

$$A_n = A_n^L \times \sin(\frac{\sigma^L}{2}) \times e^{i\theta^L/2} \times e^{i\varphi_n^L} + A_n^R \times \cos(\frac{\sigma^R}{2}) \times e^{-i\theta^R/2} \times e^{i\varphi_n^R} \quad (2)$$

where σ^L and σ^R represent the initial amplitude of RCP and LCP component, respectively, θ^L and θ^R denote the initial phases of these components. Consequently, the three focal intensities of the actual output field vary with the Poincaré parameters of the input field, mapping each HyOPB to a unique output intensity ratio $|I\rangle$. By designing a complex nonlinear mapping, we can establish a direct connection between the output intensity ratio $|I\rangle$ and the Poincaré parameter $|Q\rangle$.

The predefined amplitude and phase distributions at the focal plane are implemented using spin-multiplexed metasurfaces, facilitated by diffractive optical neural networks. During the training process, each layer of the network consists of 50×50 neuron metasurfaces with a period of 350 nm, operating at a wavelength of 532 nm. The axial distance between layers is fixed at 20 μm. For vector light field parameter mapping, increasing the number of hidden layers generally improves accuracy. However, a larger number of hidden layers typically results in a significant drop in transmittance. Therefore, we select 2 hidden layers with a total of $50 \times 50 \times 2$ neurons to achieve a fixed conversion of the predefined output field. The input light field is defined as a vortex

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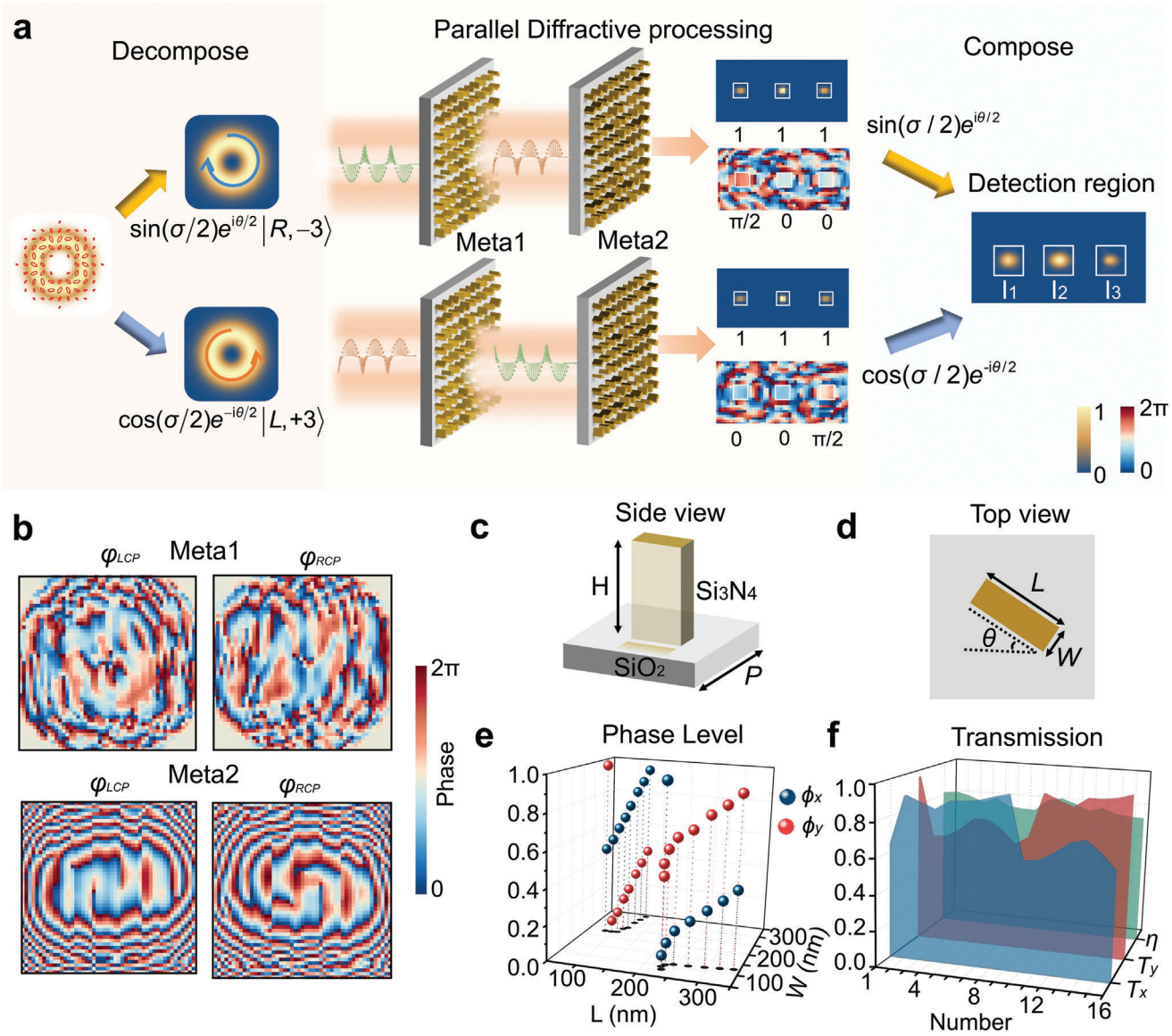

Figure 2. Working Mechanism of Spin-Multiplexed Diffractive Metasurfaces for Poincaré Parameter Detection. a) Initially, HyOPBs are decomposed into two orthogonal circularly polarized lights carrying optical vortices, and the complex amplitude coefficients of HyOPBs are closely related to the Poincaré parameters. As HyOPBs traverse the double-layer spin-multiplexed diffractive metasurface, the output becomes the complex amplitude superposition of the three predetermined focal points. b) Target phase distribution of the double-layer metasurface in the RCP and LCP channels. c,d) Illustration of the birefringent metasurface, which consists of silicon nitride nanopillars on a quartz substrate with a periodicity of P = 350nm and a height of H = 800nm. c) provides a side view of the nanopillar, while (d) offers a top view. e) The phase orders for 16 nanopillars, including ϕ_x and ϕ_y. f) The transmittance and polarization conversion efficiency of 16 nanopillars.

beam carrying a topological charge l, which is determined by the following expression:

$$LG_{l,p} = \frac{1}{w_z}\left(\frac{\sqrt{2}r}{w_z}\right)^{|l|} L_p^{|l|}\left(\frac{2r^2}{w_z{}^2}\right) e^{-\left(\frac{r}{w_z}\right)^2} e^{il\varphi} \tag{3}$$

where r and φ are the cylindrical coordinates; $w_z = w_0\sqrt{1+\left(\frac{z\lambda}{\pi w_0^2}\right)^2}$ and $w_0$represents the beam radius and $L_p^{|l|}$ is the Laguerre polynomial with the angular index l and the radial index p. We select $|R, -3\rangle$ and $|L, +3\rangle$ as the bases for HyOPBs. Under the illumination of a vortex beam, the diffractive metasurfaces show the capability of converting incident light to a designed target field **U**. The propagation of the light field follows the principles of angular spectrum diffraction theory. We define the real output light field as **O** and calculate the loss function using the mean square error (MSE)

$$loss = \frac{1}{N}\sum_{i=1}^{N}\left(|U_i|^2 - |O_i|^2\right)^2 \tag{4}$$

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The stochastic gradient descent algorithm is used to back-propagate the error and update the metasurface phase to minimize the loss function. The entire design process is illustrated in Figure S1 (Supporting Information) and is detailed in Note S1 (Supporting Information). Through iterative cyclic optimization, the final output light field converges to the designated target. The training processes for the LCP and RCP bases are conducted independently. The optimized double-layer metasurface phase distributions are shown in Figure 2b.

To independently control the phases of RCP and LCP light, an efficiently birefringent meta-atom combining geometric phase and propagation phase is employed, as shown in Figure 2c,d. The metasurface unit consists of silicon nitride (Si_3N_4) nanorods on a quartz substrate. The nanorods maintain a uniform height of $H = 800$nm, while their in-plane dimensions (L, W) and azimuth angle (θ) vary spatially to achieve the desired phase. We use the finite-difference time-domain (FDTD) algorithm to perform parameter sweeps on L and W with the defined period $P = 350$nm. More simulation details can be found in Note S2 (Supporting Information).

Based on the parameter sweep results, a set of 16 nanopillars is selected to provide 16 levels of phase, covering the full range of 2π for ϕ_x and ϕ_y, as illustrated in Figure 2e. The high transmittance (T_x,T_y) and polarization conversion efficiency (η) of the 16 nanopillars ensure the efficient operation of the diffractive metasurface at the working wavelength, as highlighted in Figure 2f. The propagation phase and geometric phase of the transmitted light are determined by ϕ_x, ϕ_y and the rotation angle θ, respectively. Thus, the required phase shift and rotation angle at each pixel are calculated based on the target phase to realize the design of the spin-multiplexed diffractive metasurfaces (the design method is shown in Note S3, Supporting Information).

3. Results and Discussion

3.1. Deep Learning-Assisted Poincaré Parameter Prediction

Figure 3a illustrates the system scheme for deep learning-driven Poincaré parameter measurement. We generated 400 sets of samples with randomly distributed Poincaré parameters $|Q\rangle$ as a training dataset and assigned them to the designed spin-multiplexing diffractive metasurface. Vector diffraction was employed to calculate the light field distribution of the output field, and the Poynting vector integral was used to determine the intensity ratio of the three focal points. Details of the additional data are available in Note S4 (Supporting Information). Through this methodology, we obtained 400 sets of Poincaré parameters $|Q\rangle$ and the corresponding intensity ratios $|I\rangle$. Once accumulating sufficient datasets, an EDNN was established to provide feature extraction and inverse mapping, so that the prediction error remains within a low range. The intensity ratio $|I\rangle$ serves as the input, and the output is the Poincaré parameter $|Q\rangle$. The EDNN comprises an input layer, four nonlinear fully connected layers, and an output layer (see Note S5, Supporting Information), which constructs the inverse mapping between $|I\rangle$ and $|Q\rangle$. This allows us to directly obtain the Poincaré parameters from the EDNN. The network training error curve of EDNN is shown in Figure 3b. This approach enables the prediction of Poincaré parameters without multiple measurements and direct characterization of the vector state of HyOPBs with a single-shot detection, which extremely simplifies the experimental process and avoids registration errors.

To clearly demonstrate the results, we randomly selected 10 sets of intensity ratios and input into the trained EDNN. The predicted Poincaré parameters closely match the target values, as shown in Figure 3c,d. The comparative results in Figure 3e,f indicate that the prediction of phase difference θ is primarily distributed along the diagonal, suggesting relatively accurate prediction results. However, several data points for the amplitude σ deviate from the diagonal. We emphasize that increasing the dataset size could enhance prediction accuracy, although this would demand significant computational resources and time.

Finally, to quantify the accuracy of the scheme, we use the mean-absolute-error (MAE) $E(Q_i) = \langle |Q_i^r - Q_i^o| \rangle$ as the test error for the i-th Poincaré parameter, where $|Q^r\rangle$ represents the actual value and $|Q^o\rangle$ denotes the output value. We first calculated the absolute errors of the Poincaré parameters, with Figure 3g,h illustrating the absolute errors of θ and σ in the test datasets. Our analysis revealed that most prediction errors are below 0.05, indicating a high level of accuracy in the results. Subsequently, we calculated the MAE for σ and θ, yielding MAE(σ) = 0.031 and MAE(θ) = 0.009. Furthermore, we calculated the average relative error of the reconstruction, finding it to be <2.68% for σ and <1.84% for θ. These findings further demonstrate that our scheme provides reliable and accurate predictions.

3.2. Single-Shot Characterization of HOPBs

As proof of concept, we first demonstrated the single-shot characterization of higher-order Poincaré beams (HOPBs). HOPBs, a more common form of HyOPBs, are distinguished by opposite topological charges between LCP and RCP optical vortices.[57,58] Most research focuses on the generation of HOPBs,[59–63] while detection is often overlooked, with many studies still relying on the rudimentary method of using rotating polarizers.

Building upon the proposed method, we designed a dual-layer spin-multiplexing diffractive metasurface and trained the associated EDNN model to achieve single-shot characterization of HOPBs ($|R, -3\rangle$and$|L, +3\rangle$). We performed vector simulations to verify the accuracy of the single-shot characterization scheme. Specifically, we randomly selected three distinct modes A (σ = 0.783, θ = 1.662), B (σ = 1.576, θ = 5.465), and C (σ = 2.347, θ = 1.472). from the higher-order Poincaré sphere, as illustrated in **Figure 4a**. These vector modes serve as the input, incident on the dual-layer diffractive metasurface, while a monitor simultaneously records the intensity distribution at the output plane. The input modes are depicted at the top of Figure 4b. The vector simulation is employed to calculate the far field. The middle of Figure 4b presents the electric field distribution of the output layer for three selected vector modes, where the intensity ratios of the three focal points display clear differences.

Subsequently, we use the Poynting vector integral to calculate the energy ratios of the three focal points, which are then normalized before being fed into the trained EDNN model. The EDNN efficiently extracted the Poincaré parameters of the input vector modes. Using these predicted Poincaré parameters, we

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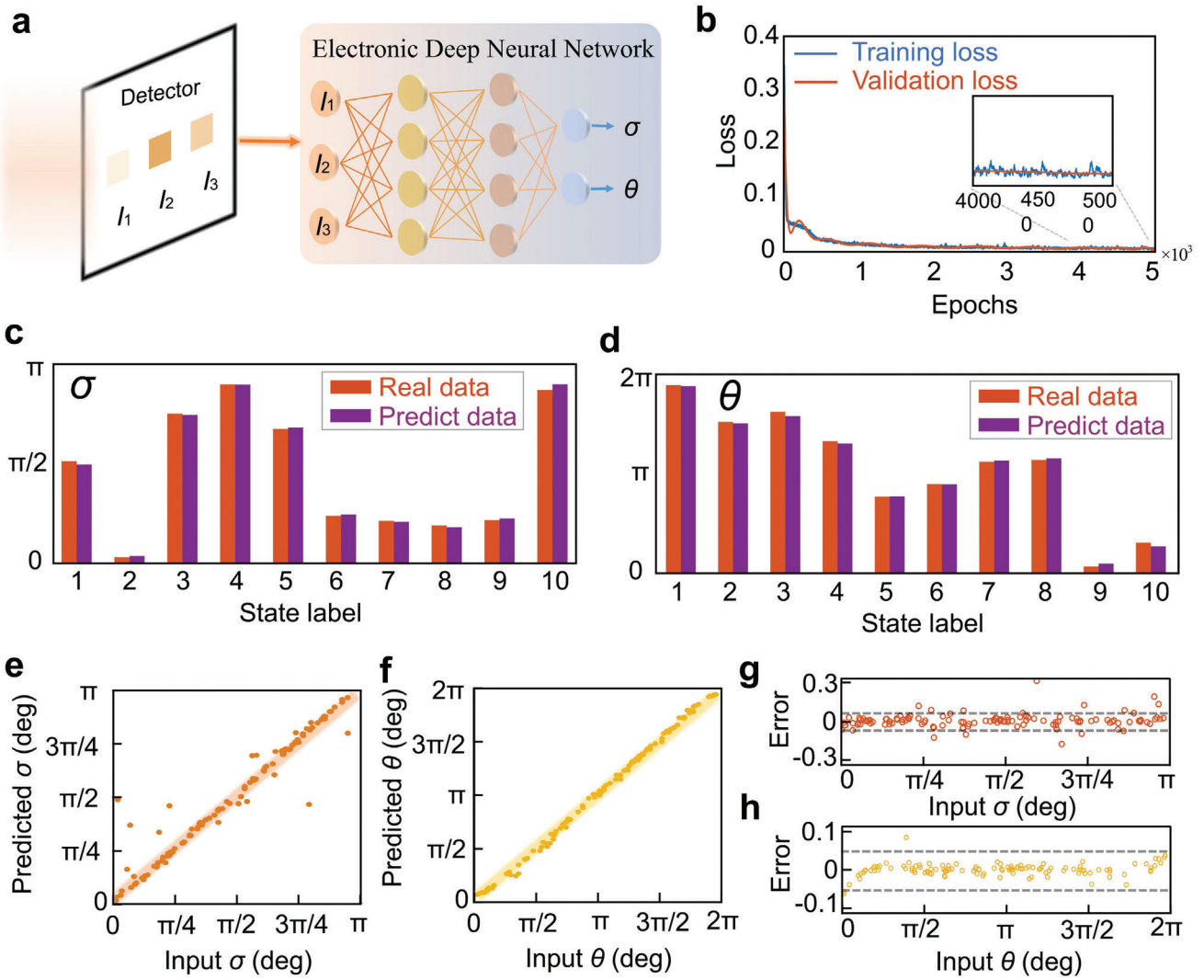

Figure 3. Poincaré Parameter Prediction Supported by Deep Learning. a) The scheme of Poincaré parameter prediction. b) Training error curve of the EDNN. c,d) Comparison of predicted results with actual outcomes for 10 randomly selected test datasets, including the phase difference θ and amplitude ratio σ. e,f) Comparative results of Poincaré parameters from 100 sets of predicted values and actual values. Data points inside the diagonal are considered correct predictions. g,h) Absolute error of the test data. The absolute error is defined as the difference between the output and the corresponding real value.

reconstructed the three vector modes, with the results displayed at the bottom of Figure 4b. Apparently, the reconstructed vector patterns closely match with the input test patterns, demonstrating the effectiveness of our proposed single-shot characterization scheme for HOPBs.

To further assess the robustness of our approach, we randomly selected 15 different modes to test the reconstruction ability of our scheme. Notably, these modes do not belong to the original 100 test sets, confirming the generalizability of our proposed single-shot characterization scheme. A comparison between the reconstructed and theoretical results reveal that the reconstruction errors are ±2.35% for σ and ±1.54% for θ. Evidently, our scheme clearly outperforms traditional Stokes polarization detection in terms of reconstruction accuracy, compact integration as well as real-time detection.

3.3. Single-Shot Characterization of HyOPBs

Finally, to demonstrate the capability of the characterization of arbitrary vector beams, we utilize the proposed single-shot characterization scheme to detect HyOPBs. Compared to HOPBs, the LCP and RCP optical vortices in HyOPBs possess completely different topological charges, leading to a more complex transverse polarization distribution in the combined vector beam. As shown in **Figure 5a**, we present a HyOPBs characterized by the basis $|R, -5\rangle$and $|L, +3\rangle$. It is evident that the lack of symmetry in the optical vortices carried by LCP and RCP results in a rapid variation of the polarization state distribution in the transverse plane, which poses significant challenges for directly reconstructing the polarization state to predict the Poincaré parameters.

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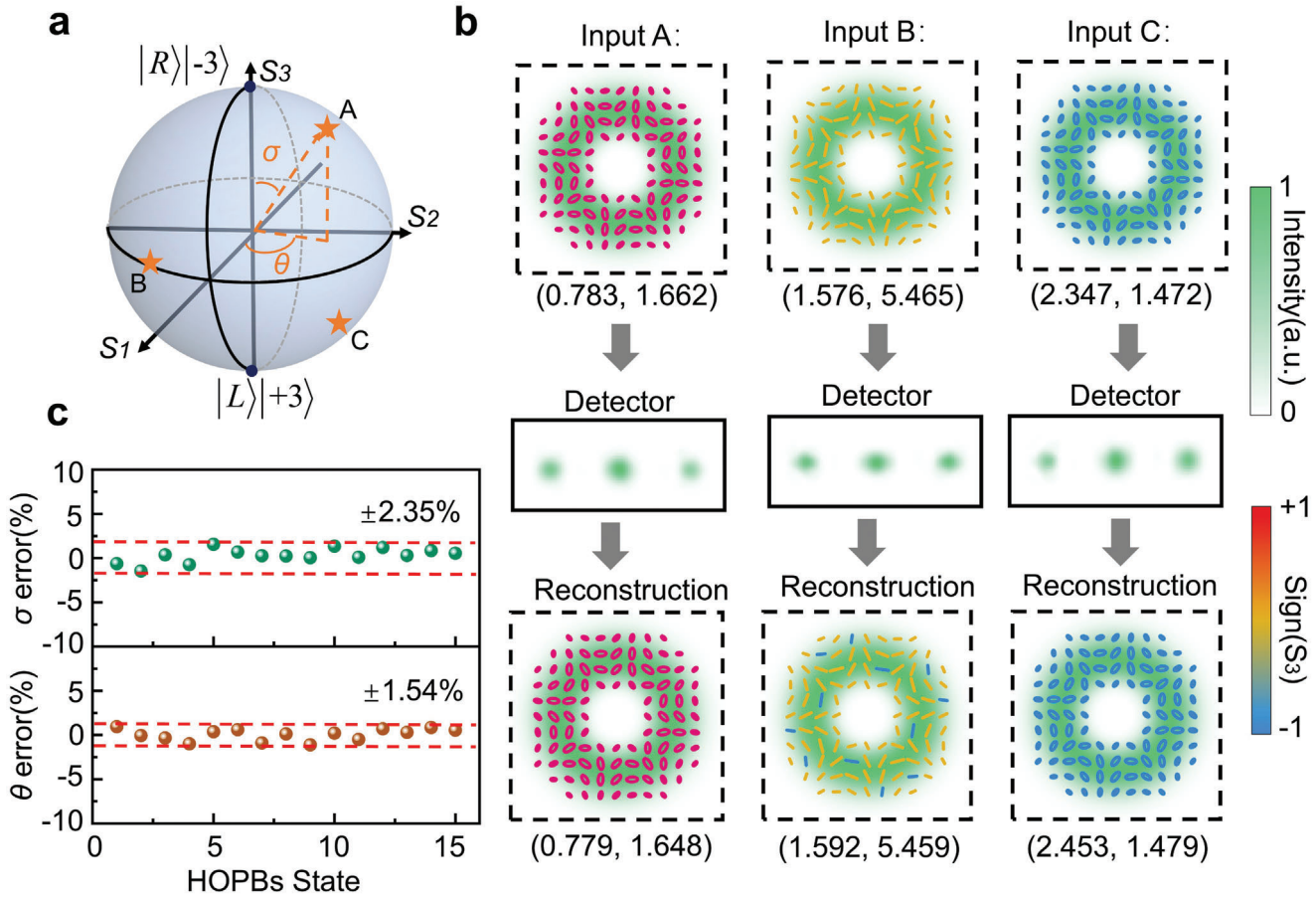

Figure 4. Verification of Single-Shot Characterization of HOPBs. a) HOPBs with basis of $|R, -3\rangle$ and $|L, +3\rangle$. b) Reconstruction results of three randomly selected input vector modes. The top row shows the theoretical distribution of the three modes, the middle row shows the output intensity after diffractive metasurface processing, and the bottom row shows the vector modes predicted by the EDNN. c) Prediction error results for 15 randomly selected modes.

Fortunately, we can employ a dual-layer spin-multiplexed diffractive metasurface to manage complex polarization distributions and directly extract the Poincaré parameters of the incident light field without the need for polarization reconstruction. We utilize two optical vortices with different topological charges as inputs to design the diffractive metasurfaces. Subsequently, we perform vector simulations and Poynting vector integration to calculate the output energy ratio after the beams pass through the metasurface, normalizing this ratio before inputting it into the EDNN model. A total of 400 datasets were used to train the corresponding EDNN model. Next, we randomly selected three distinct patterns from the hybrid-order Poincaré sphere: A ($\sigma = 1.185$, $\theta = 1.662$), B ($\sigma = 1.576$, $\theta = 5.462$), and C ($\sigma = 2.867$, $\theta = 1.663$). Figure 5b illustrates the results of the characterization and reconstruction, demonstrating that the reconstructed HyOPBs closely agree with the input vector modes. These results further validate the effectiveness of our proposed single-shot characterization scheme in universal characterization and reconstruction of HyOPBs.

Finally, we evaluate the robustness of the proposed dual-layer spin-multiplexing diffractive metasurface system. Among the various influencing factors, the lateral displacement between the two metasurfaces has a decisive impact on vector mode processing performance. Additionally, variations in the gap distance lead to a significant drop in system performance. A more detailed discussion is provided in Note S6 (Supporting Information). Notably, the technical challenges associated with system alignment can be effectively addressed by implementing an end-to-end global design flow that fully accounts for potential experimental errors. In terms of fabrication, photoresist can be used as a spacer layer for multilayer integrated cascade metasurfaces. This method achieves interlayer self-alignment during manufacturing,[64–66] eliminating the need for subsequent manual alignment of the single-piece diffraction metasurface and significantly improving both preparation efficiency and accuracy.

4. Conclusion

In conclusion, we proposed and successfully validated a single-shot characterization scheme for HyOPBs using a double-layer spin-multiplexed diffractive metasurface. By integrating diffractive optical neural networks with electronic deep neural network models, our approach addresses the limitations of traditional Stokes parameter measurements, allowing for real-time characterization and reconstruction of HyOPBs. In contrast to conventional methods that depend on multiple polarization

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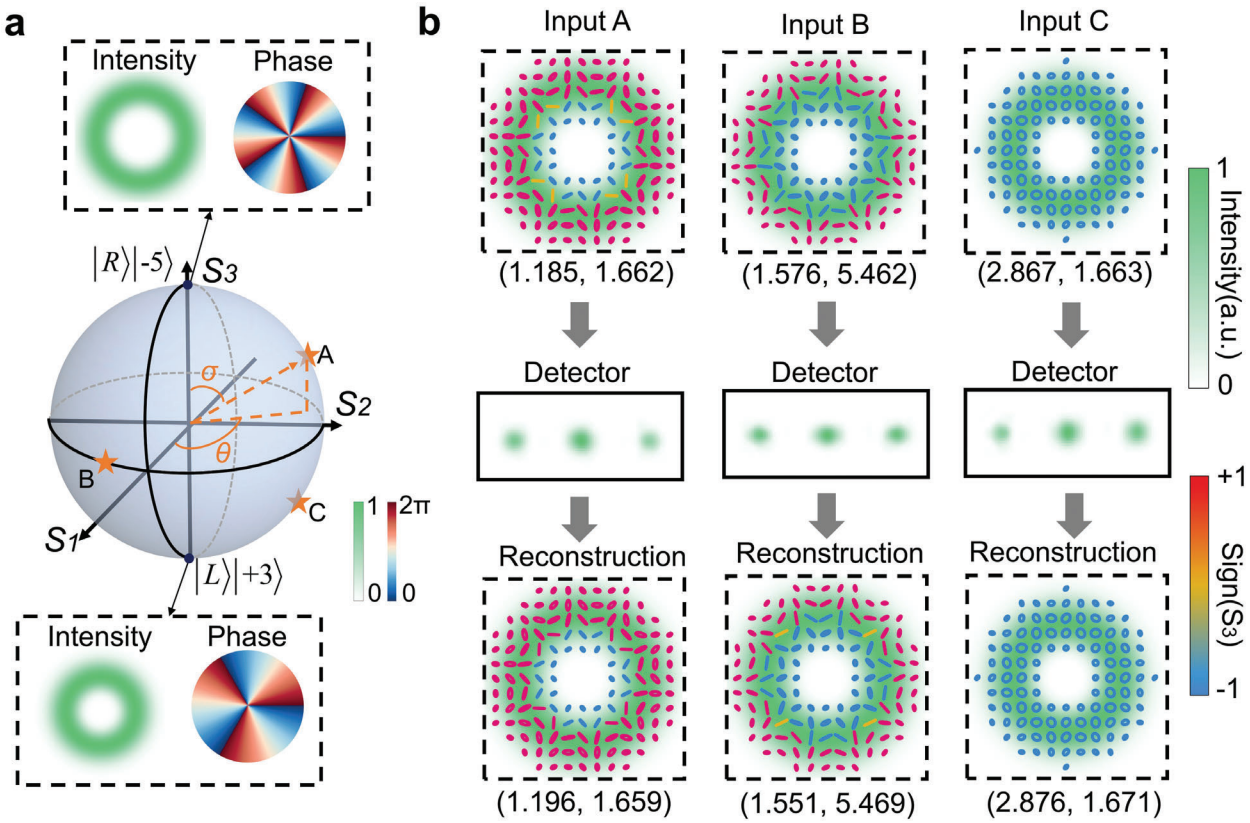

Figure 5. Verification of Single-Shot Characterization of HyOPBs. a) HyOPBs with basis of $|R, -5\rangle$and $|L, +3\rangle$. The insets are the phase and intensity results of the basis modes, respectively. b) Reconstruction results of three randomly selected input vector modes. The top row shows the theoretical distribution of the three modes, the middle row shows the output intensity after diffractive metasurface processing, and the bottom row shows the vector modes predicted by the EDNN.

projections, our scheme fully characterizes the vector properties of HyOPBs using only a single intensity measurement, marking a significant departure from traditional Stokes-based schemes. Furthermore, our scheme is adaptable to arbitrary vector Poincaré sphere beams, such as perfect Poincaré beams and higher-dimensional vectorial structured light. Our work enables rapid, accurate, and single-shot characterization of HyOPBs in a compact integrated optical sensor system, opening new avenues for advancements in high-capacity optical communications, entangled quantum information, and high-dimensional optical encryption.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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diffractive metasurfaces, diffractive optical neural networks, hybrid-order Poincaré beams, single-shot characterization

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